Soumya Sharma

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Education

Birla Institute of Technology, Mesra (Patna Campus) — Patna, India
Bachelor of Technology in Electronics and Communication Engineering; CGPA: 7.2/10 — *2024 – May 2028*

Children College, Azamgarh — *Class XII (ISC): 93% | Class X (CISCE): 92.4%* — Azamgarh, India

Experience

NS Apps Innovation — Sep 2026 – Nov 2026
AI/ML & React Developer Intern — *India*

- Built a hybrid **OKF + RAG** chatbot answering Bihar MSME policy questions with citations: a Google **Open Knowledge Format** graph (249 concepts, 311 edges) answers exact facts in **0.02–0.13 s vs. 20–110 s** via the LLM; open questions go through **BGE-M3** dense + sparse retrieval in **Qdrant**, rank fusion and a **bge-reranker-v2-m3** cross-encoder, with a coverage check that declines low-evidence answers. **FastAPI** + React; 109 figures source-verified; 51 tests
- Engineered the backend of **Yoda**, an AI desk robot, on an **NVIDIA DGX Spark**: an asyncio **FastAPI** server queues JPEG frames (202, drop-oldest) and holds an authenticated **WebSocket** for commands and 16 kHz audio; a PyTorch-free **ONNX Runtime** chain (RTMO-s → YuNet → SFace → FER+) runs on CPU at **60–70 ms/frame, 5 FPS, 0 dropped frames**, with keypoint wave detection and open-set face ID
- "Hey Yoda" voice loop: **Silero VAD** endpointing (0.3 s pre-roll, 0.7 s pause) → int8 **faster-whisper** (benchmarked base/small/large-v3-turbo; small: 1.4 s per 2.5 s of speech on CPU) → fuzzy wake match → **Qwen3-VL-8B** on the live frame; unaddressed speech is dropped in memory; a Pydantic allow-list gates commands to hardware; 167 tests

Undergraduate Research Intern – Nanopore Biosensor Signal Processing — Aug 2026 – Present
Under Dr. Naren Das, Dept. of ECE, BIT Mesra — *Patna, India*

- Built a 17-module Python pipeline (NumPy, SciPy, scikit-learn) and curated a **440-recording (3.1 GB)** dataset; the event classifier reached **0.84 balanced accuracy** on an independent test day vs. 0.55 for a cross-session baseline

Projects

Amazon ML Challenge 2026 – Entity Resolution | *Python, LightGBM, PyTorch, Transformers* — GitHub

- Led a 3-member team matching 1.7M businesses against ∼**10M noisy records**: multi-key blocking + a two-stage **LightGBM** cascade (73 features) cut candidates from ∼300 to **5.8 per entity** at 0.979 recall; a multilingual **mmBERT-small (140M)** cross-encoder fine-tuned on Kaggle GPUs re-scores uncertain pairs (AUC 0.955 vs. 0.931). Final **public-leaderboard F0.5 of 0.968** among ∼90,000 registrants

Twigg – LLM Copilot (Browser Extension) | *JavaScript, Manifest V3, Gemini API, Ollama* — GitHub

- **Manifest V3** side-panel copilot: rates roles against your resume, autofills forms (never submits) and drafts tailored bullets and outreach; an **LLM guardrail** rejects rewrites that add unsupported numbers or tools

Technical Skills

Languages & Frameworks: Python, C/C++, TypeScript, JavaScript, SQL, React
ML / DL: PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, ONNX Runtime, OpenCV, Pandas, Polars
LLM / GenAI: RAG, OKF knowledge graphs, Qdrant (vector DB), embeddings, rerankers, Ollama, Whisper, VLMs
Backend: FastAPI, asyncio, Pydantic, pytest, SQLAlchemy, WebSockets, React, Next.js, Git, Linux

Applied Patent and Publications

- **Hybrid Psychometric-Machine Learning Platform for Multi-Modal Mental Health Screening of Engineering Students**: IRT-based adaptive screening; built its FastAPI + Next.js platform (**92 backend tests**) with consented counsellor reports and k-anonymous aggregates

Achievements & Leadership

- **Finalist**, TantraFiesta 2025 (IIIT Nagpur) and Coinbase-sponsored Web3 Hackathon (BIT Mesra); Round 1, **Adobe Hackathon 2026**
- **Mentoring**: conducted ML workshops and tutored ∼300 students at BIT Mesra
- **Head of Publicity, TECHNIKA 26-27** (∼2,000 participants); **Ignite** Designing & Drafting Team
- **Basketball** gold, 2× inter-branch and inter-college; guitarist, Pratibimb Society (**1st prize**, band competition)